



PH142 Review: Week 2

PH142 GSI team
Summer 2026



Announcements

Due Dates:

- 7/15: Quiz08
- 7/16: Midterm 1
- 7/17: Quiz09, Data Project Part 1



Midterm 1 Details

- **Date:** Thursday, July 16th
- **Time:** Available for 24 hours (12:00AM - 11:59PM)
- **Duration:** 60 minutes, timed
- **Material Covered:** Lectures 1-8, Labs 1-3
- **What to Expect:**
 - Similar format to quizzes, exam will be taken via Gradescope
 - Mix of multiple choice, select all, and fill-in-the-blank
 - You may use one page (front & back) of handwritten or printed notes
 - You may not use the internet or AI tools to complete the exam



Key Points

- PPDAC
 - What does this stand for?
- Types of Problems
 - What is the difference between descriptive, causal, and predictive problem types?
- Describing Data
 - What kind of observations do we have? (e.g. Individual people, lab tests, etc.)
- What kind of variables do we have?
 - Nominal, Ordinal, Continuous, Discrete?



PPDAC Framework

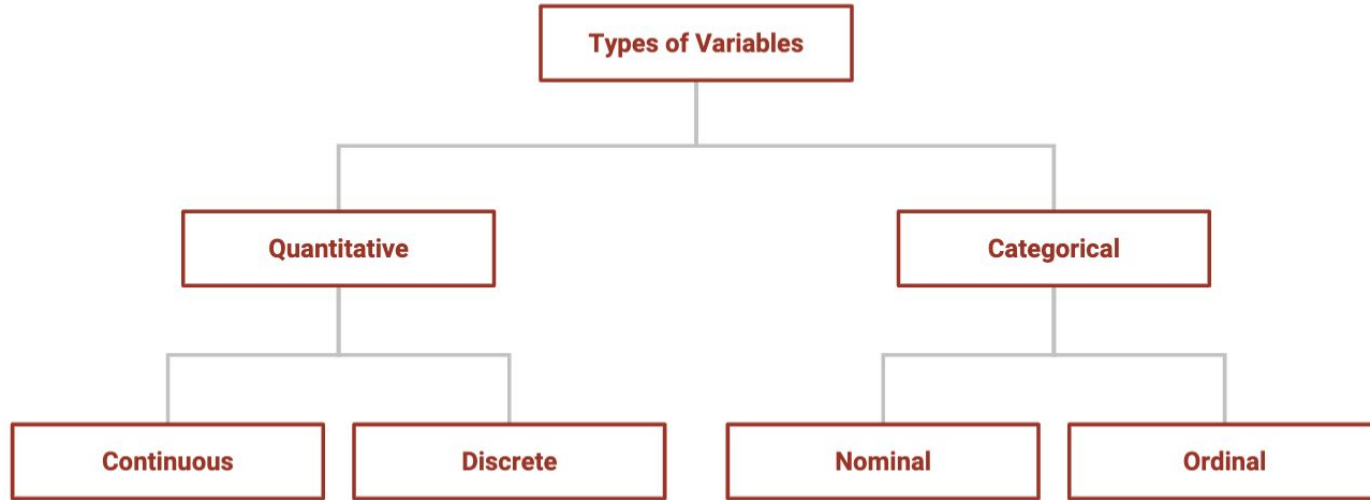
- **Problem:** A clear statement of what we are trying to achieve.
- **Plan:** The procedures we use to carry out the study.
- **Data:** The data which is collected according to the Plan.
- **Analysis:** The data is summarized and analysed to answer the questions posed by the Problem.
- **Conclusion:** Conclusions are drawn about what has been learned about answering the Problem.



Problem Types in PPDAC

- **Descriptive:** Learning about some particular attribute of a population
 - Example: Trying to determine the mean amount that grad students pay for rent at Berkeley
- **Causative/Etiologic:** Do changes in an explanatory variable cause changes in a response variable?
 - Example: If we change the stipend amount for grad students, how does that change amount of money grad students at Berkeley choose to spend on rent?
- **Predictive:** How can we best predict the value of the response variable for an individual?
 - Example: Can we identify some explanatory variables that help predict the amount of money that grad students choose to spend on rent? Note that these variables may not be causing the grad students to change the amount of rent they spend. It may simply be an association relationship rather than a causal relationship.

Types of Variables



Can you give an example for each of these variable types?



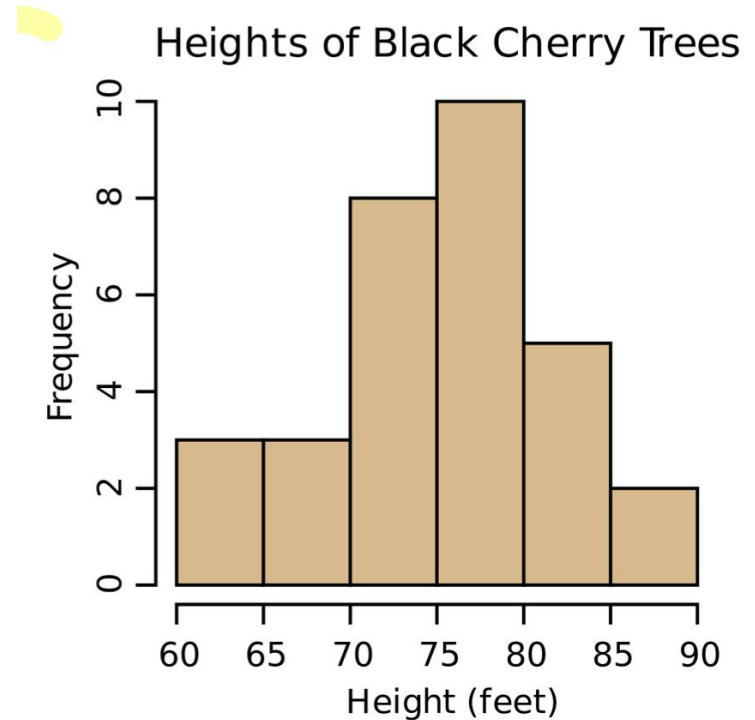
Visualizing and Describing Data

- Bar Charts vs. Histograms
 - What does the height of a bar represent? Frequency (count)
 - What does the x axis represent? Levels of the variable
- What measures of central tendency did we learn?
 - Which are resistant to outliers?
- What measures of spread did we learn?
 - Which are resistant to outliers?

Plots: Histograms

Histograms are used to illustrate the distribution of a numeric (continuous or discrete) variable. There are no spaces between bars.

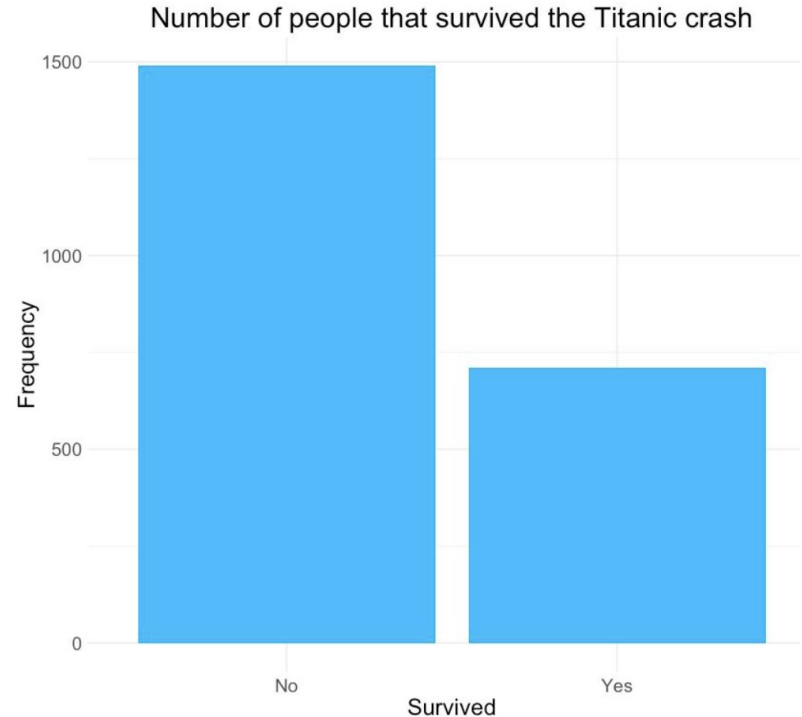
*Note that it does not make sense to rearrange the bars of a histogram



Plots: Bar plots

Bar plots are used to illustrate the distribution of a categorical variable, with spaces between bars

- The “distribution” of a categorical variable is the count or percent of individuals in each category





Continuous vs. Categorical Data Visualization

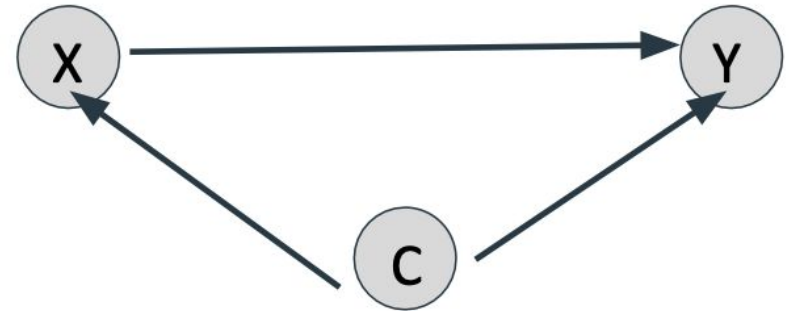
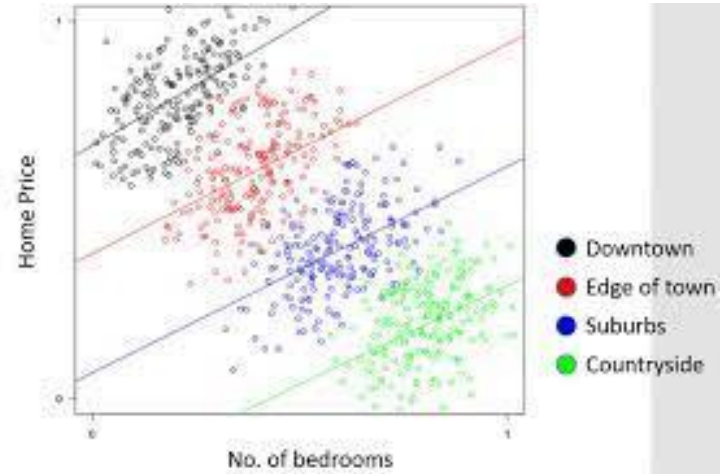
For two **continuous** variables?

- Scatterplots, correlation coefficients, linear regression

For two **categorical** variables?

- Bar charts, tables, conditional v. marginal distributions, think about Simpson's Paradox

Simpson's Paradox

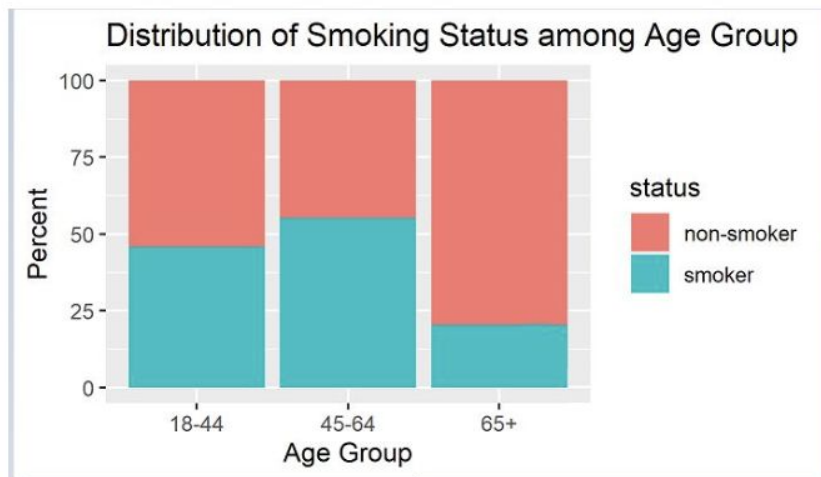


Visual Summaries in R

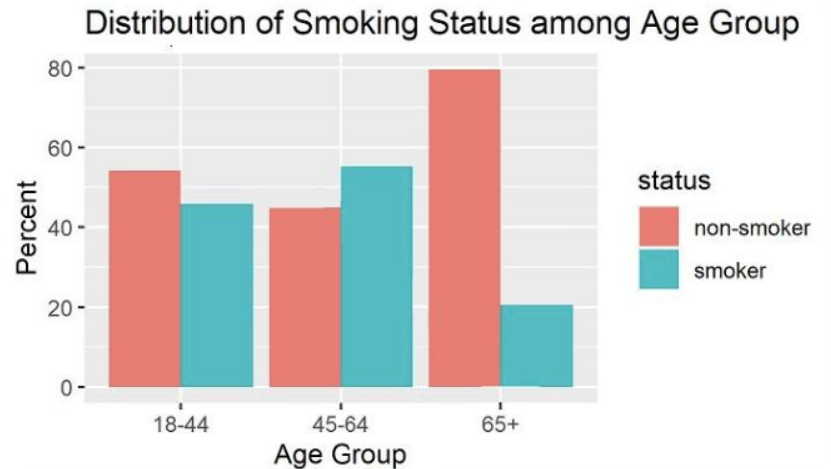
- **ggplot** - `new_dataset <- dataset %>% ggplot (aes(x= variable, y=variable))`
- **Histograms** - `geom_histogram;`
 - `ggplot(data = dataset, aes(x = variable)) +
 geom_histogram(col = "white", binwidth = 10) +
 labs(x= "", y = "") +
 theme_minimal(base_size = 15)`
- **Bar plots** - `geom_bar(stat='count')` or `geom_bar(stat='identity');`
 - `ggplot(dataset, aes(x= variable, y=variable)) +
 geom_bar(stat = "identity") +
 labs(y = "", x = "") +
 theme_minimal(base_size = 15)`
 - Stat = "identity" tells geom_bar/histogram that we supplied a y variable this is exactly what we want to plot and we do not need geom_bar to calculate the number or percent for us
 - Stat = count will tell ggplot to calculate percentages
- **Dodged bar plots** - `geom_bar(aes(x=var, fill=var), stat='identity' , position='dodge')`
 - Position = "dodge" → puts bars next to each other
 - Position = "stack" → stacks variables on top of each other in one bar
 - `aes(fill = var)` links the bar's color fill to the x variable
 - `Fct_relevel` = allows you to line up the bars in ascending order

Stacks vs. Dodged Bar Chart

Using the previous example, let's visualize the data with a stacked and dodged bar chart:



```
ggplot(data = my_data, aes(x = age, y = percent)) +  
  geom_bar(aes(fill = status), stat = "identity",  
  position = "stack")
```



```
ggplot(data = my_data, aes(x = age, y = percent)) +  
  geom_bar(aes(fill = status), stat = "identity", position =  
  "dodge")
```



Quantitative Summaries and IQR

- **Quantitative Summaries**

- **dplyr** - `dataset <- dataset %>% dplyr_function(argument_1 = , argument_2 = )`
- **mutate** - `mutate(new_column = 500 * existing_column)`
- **summarize** - `summarize(summarize_stat_name = 500 * existing_column)`
 - Note: see next slide for more on DPLYR library

- **Interquartile Range (IQR) Q3 - Q1**

- Resistant to outliers
- Quantiles using R
 - `Dataset %>% summarize(Q1 = quantile(var, 0.25) , median = median(var) , Q3 = quantile(var, 0.75))`
 - *To get exact answer as you would by hand = `quantile(data, 0.25, type = 2)`

dplyr library

- Functions to know
 - Filter
 - Mutate
 - Summarize / group_by
 - Arrange
 - Select

Libraries at A Glance: dplyr

Managing large data frames is hard – dplyr can help! With dplyr, you can organize your data to match your needs! Here are some useful functions to help you transform your data frame from scattered to focused.

filter

This function allows you to “filter” through your data frame and select observations that meet your set criteria through logical statements (`==`/`>`/`<`/`!=`).

```
filtered_data <- filter(data file, condition)
filtered_data <- filter(pet, state == "CA")
```

Pets owned in USA			
State	Cat	Dog	
CA	0	1	
AZ	0	1	
CA	1	1	
DE	2	0	

Pets owned in USA			
State	Cat	Dog	
CA	0	1	
CA	1	1	

mutate

This function allows you to change variables or append columns to a data frame. This can be useful to perform new calculations on existing data.

```
mutated_data <- filter(data file, condition)
filtered_data <- mutate(pet, Total_Pets = Cat + Dog)
```

Pets owned in USA			
State	Cat	Dog	
CA	0	1	
AZ	0	1	
CA	1	1	
DE	2	0	

Pets owned in USA				
State	Cat	Dog	Total Pet	
CA	0	1	1	
AZ	0	1	1	
CA	1	1	2	
DE	2	0	2	

summarise/group_by

This function allows you to complete single number output calculations on your entire dataset (i.e. means, total sums, etc.). You can use the `group_by` function to arrange summary functions by category. The `group_by` function does not change the data frame.

```
summary_data <- summarise(data file, condition)
summary_data <- summarise(pet, mean_cats = mean(Cat), mean_dogs = mean(Dog))
group_data <- summarise(data file, var)
group_data <- group_by(pet, State)
summarise(group_data, mean_cats = mean(Cat),
```

Pets owned in USA			
State	Cat	Dog	
CA	0	1	
AZ	0	1	
CA	1	1	
DE	2	0	

mean_cats		mean_dogs	
<dbl>		<dbl>	
0.75		0.75	
1 row			

State	mean_cats	mean_dogs
<chr>	<dbl>	<dbl>
AZ	0.0	1
CA	0.5	1
DE	2.0	0

arrange

This function allows you to arrange data numerically or alphabetically by variable.

```
arrange_data <- arrange(data file, var)
arrange_data <- arrange(pet, State)
```

Pets owned in USA			
State	Cat	Dog	
CA	0	1	
AZ	0	1	
CA	1	1	
DE	2	0	

Pets owned in USA			
State	Cat	Dog	
AZ	0	1	
CA	0	1	
CA	1	1	
DE	2	0	

select

This function allows you to select variables to include or exclude in your data frame. Use the “-” sign to exclude.

```
select_data <- select(data file, var)
select_data <- select(pet, -Dog)
```

Pets owned in USA			
State	Cat	Dog	
CA	0	1	
AZ	0	1	
CA	1	1	
DE	2	0	

Pets owned in USA			
State	Cat		
AZ		0	
CA		0	
CA		1	
DE		2	

Useful Tip: The dplyr library includes a `%>%` (pronounced “pipe”) operator function. You can use it to “chain” the functions above to succinctly complete more than one function on one dataset at the same time. This makes it easy to add steps and minimize local variables stored in the environment.

Variance, SD, and 5 Number Summary

Sample Variance

- s^2 represents sample variance
- $s^2 = 1/(n-1) \sum (X_i - \bar{X})^2$
- Use `summarize(dataset %>% summarize(name_var = var(variable)))`

Sample Standard Deviation

- s represents sample standard deviation
- Square root of sample variance
- Use `summarize(dataset %>% summarize(name_sd = sd(variable)))`

Five Number Summary = Min, Q1, median, Q3, max

- To visualize, use a box plot
 - Center line = median
 - Top box = Q3
 - Bottom box = Q1
 - Top whisker = Max or $Q3 + 1.5 \cdot IQR$
 - Bottom whisker = Min or $Q1 - 1.5 \cdot IQR$
- Boxplots in R = Use `ggplot()`'s `geom_boxplot()`
- Use `dplyr`'s `summarize()` function to calculate 5 number summary
- Ex. `dataset %>% summarize(min = min(var) , Q1 = quantile(var, 0.25) , median = median(var) , Q3 = quantile(var, 0.75), max = max(var))`

Scatterplots



Scatterplot - `geom_point` is used for scatterplot

Ex. `name of plot <- ggplot(data = dataset, aes(x=var, y=var)) +
 geom_point(na.rm = TRUE) + theme_minimal(base_size =
 15) +
 labs (x= "", y= "", title= "")`

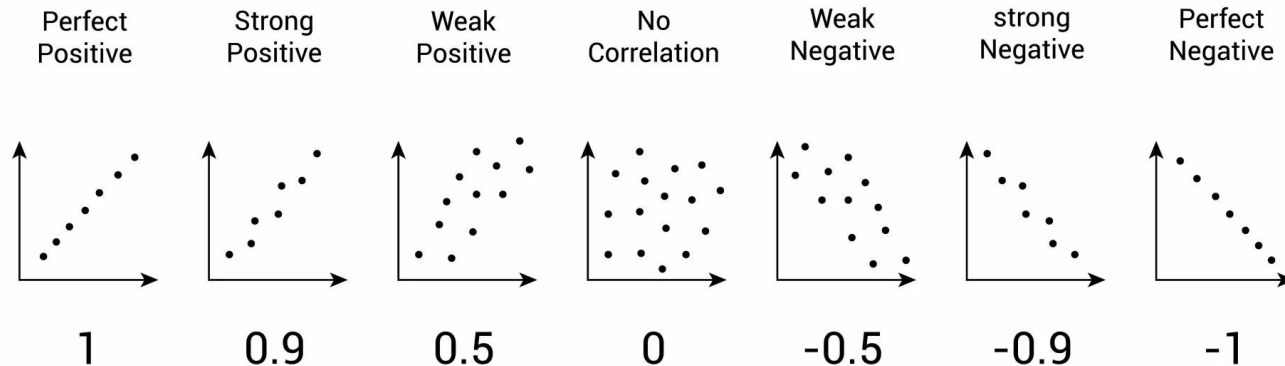
- To color the points by gender include `col=gender`

Ex. `name of plot <- ggplot(data = dataset, aes(x=var, y=var)) +
 geom_point(aes(col=gender))`

- To create separate plots for combinations of levels of 2 vars i.e. (gender) use `facet_wrap` ex. `facet_wrap ( ~ gender)`

Correlation

- Correlation measures the **strength** and **direction** of a linear relationship between two quantitative variables
- Also written as r
- Takes values between -1 to 1 inclusively



Regression

- **Regression:** Straight line fitted to data to minimize distance between the data and fitted line
 - “Line of best fit” = $a + bx$
 - a = **intercept** (Predictive Value of y when $x=0$)
 - b = **slope** $r^*(s_y/s_x)$
- Interpretation
 - Intercept : the value of the outcome when $x = 0$
 - Slope: For a one-unit change in x , the outcome changes by [number] [units]

$$y = a + bx$$



Regression Cont.

- `lm()` is the function for a linear model
 - The 1st argument that `lm()` wants is a formula $y \sim x$ (Y = response X = explanatory)
 - The 2nd argument is the data set
 - `lm(formula = y ~ x, data = your_dataset)`
 - Add regression line to a scatterplot using `geom_abline(slope=, intercept=)`
- **Interpretation of `lm()`:** A one unit change in **X** is associated with a ____ increase/decrease of **Y**.
- **Interpretation of `r-squared`:** the fraction of the variation in the values of y that is explained by the line of best fit (the regression of y on x)
-



Regression Cont.

- **Correlation (ρ or r)**

- Degree to which two variables move in coordination with one another
- $-1 < \rho < 1$

$$\frac{\text{Cov}(x,y)}{\text{SD}(x) \text{SD}(y)}$$

- **Regression Coefficient (β)**

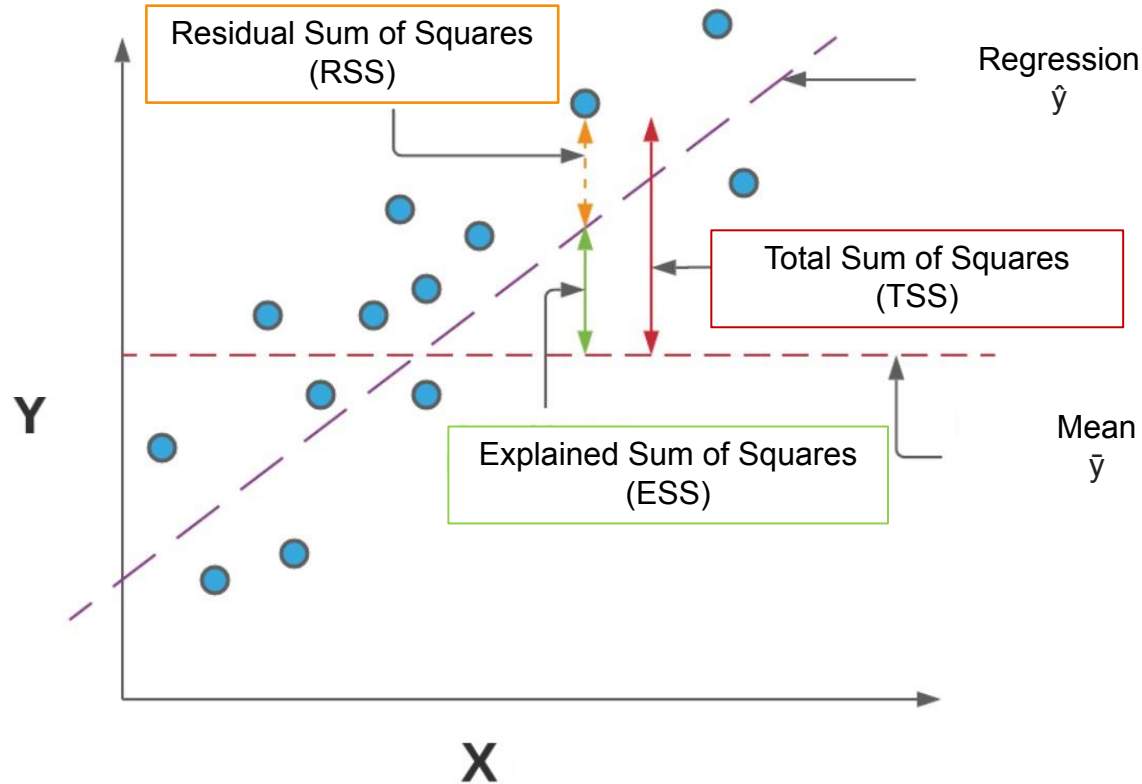
- Change in the outcome variable for every 1-unit of change in the predictor variable

$$\frac{\text{Cov}(x,y)}{\text{Var}(x)}$$

Regression Cont.

- **R^2 (of a linear model)**
 - $0 < R^2 < 1$, also ρ^2
- Goodness of fit of the fitted regression line to a set of data
 - $R^2 = ESS / TSS = 1 - RSS / TSS$
- **Total Sum of Squares = Explained SS + Residual SS**
 - $TSS = ESS + RSS$
 - $\sum (y - \bar{y})^2 = \sum (\hat{y} - \bar{y})^2 + \sum (y - \hat{y})^2$

Regression Cont.



Key Terms



Counterfactual - (how long would a person have lived without treatment X? Aka The “Ideal Experiment”); Observed: Person A $\rightarrow$ exposure to x; Unobserved: Person A $\rightarrow$ no exposure to x
**the counterfactual

Confounding - a relationship between your variable of interest and your outcome of interest is confounded when there is a variable that is associated with both the exposure and the outcome, and is not on the causal pathway between the two

Conditional distribution - distribution of variable within or conditional on the level of a second variable

Marginal distribution - “in the margin” of the table; row total or column total are the two margins of a two-way table $\rightarrow$ answers ?’s about the overall distribution of one variable

Simpson’s Paradox - An association or comparison that holds for all of several groups can reverse direction when the data are combined to form a single group. This reversal is called Simpson's Paradox

Observation vs. Experimentation



- A study is observational if the researcher **observes** what happens and does not control who is treated or exposed.
 - Does not control for confounding
- A study is experimental if the investigator is **experimenting** (or intervening) by controlling who is treated or exposed.
 - In an experimental study, the exposure is assigned by a randomization mechanism that is controlled by the investigator

Population, Sampling, and Conditional Selection

Population of Interest

- Target Population - entire group of individuals about which we want estimates to apply *problem in PPDAC*
- Study Population - part of the population which we can select individuals & collect information to draw conclusions about the entire population

Sampling

- Simple Random Sample - sample chosen by chance, where each individual had the same chance of being selected
- In R an SRS of size 100 can be generated by: `name_1 <- dataframe %>% sample_n(100)`

Conditional selection

- Conditional on exposure or outcome
- If we are choosing people to participate in our study based on their exposure status this is generally a **cohort design**
- If we are choosing people to participate in our study based on outcome status this is generally a **case control design**
- If we select participants conditional on exposure OR outcome then marginal distribution of exposure is not meaningful

Sources of Bias



- **Sampling bias** if sampling frame does not cover target population aka “undercoverage bias”
- Participation once sampled an individual may not agree to participate -- referred to as “**response**” or “**nonresponse**” bias participants lost to follow-up
- **Contamination** if people who were randomized to control receive the medication or exposure that was intended to be given only to treatment arm participants
- **Adherence** once a participant is randomized to a given treatment, they do not always adhere to that treatment
- Measurement Error
- **Self-report bias** associated w/ willingness or ability to report info = social desirability bias + recall

Participation Activity!

Designing a Mini-Study (5 minutes)

In groups of 4-5, design a small public health study on a population of your choice! Answer the questions below and upload (text-entry) into Gradescope for participation credit.

1. Problem: What is your study trying to achieve?
2. Problem Type: Is your problem predictive, causal, descriptive?
3. Name one explanatory and one response variable. Label each as continuous, discrete, nominal, or ordinal.
4. Identify one possible confounder and how it could affect your study.
5. Identify one source of bias in your study (ex. sampling bias, response bias, coherence, measurement error) and how it could be addressed.

One group member will upload a text-entry response into Gradescope. Then, add the names of your group members to the submission to receive participation credit.

Practice Problems

Question 1

Adapted from “Effects of water quality, sanitation, hand washing, and nutritional interventions on diarrhea and child growth in rural Bangladesh: a cluster randomized controlled trial”

Diarrhea and growth faltering in early childhood are associated with subsequent adverse outcomes. The authors aimed to assess whether water quality, sanitation, and hand washing interventions alone or combined with nutrition interventions reduced diarrhea or growth faltering. There were multiple intervention condition, one of which was a nutritional intervention only. One of the outcomes was childhood stunting.

The authors present the following information:

Group	Stunting	No Stunting	Total
Nutrition	186	381	567
Control	451	652	1103
Total	637	1033	1670

1a.) What is the marginal probability of stunting in this table?

1b.) What are the probabilities of stunting conditional on study group?

1c.) What kind of variable is stunting?

Question 2



Look at the title of this article from New England Journal of Medicine 2019; 380:415-424

"Partial Oral versus Intravenous Antibiotic Treatment of Endocarditis a randomized, noninferiority, multicenter trial"

Which of the following study design terms apply to this study:

- A. Experimental
- B. Observational
- C. Simple Random Sample
- D. Case-Control

Question 3

We have a dataset that looks like this:

```
## # A tibble: 4 x 4
##   smoking    lung_cancer percent number
##   <chr>      <chr>          <dbl> <dbl>
## 1 smoker    lung cancer      4.8    12
## 2 smoker    no lung cancer   95.2   238
## 3 non-smoker lung cancer      0.9     7
## 4 non-smoker no lung cancer  99.1   743
```

Which of these statements would produce our preferred bar graph for comparing lung cancer by group

- A. `geom_bar(aes(fill = lung_cancer), stat = "identity", position = "stack")`
- B. `geom_bar(aes(fill = lung_cancer), stat = "identity", position = "dodge")`
- C. `geom_bar(aes(fill = lung_cancer), stat = "count", position = "stack")`
- D. `geom_bar(aes(fill = lung_cancer), stat = "count", position = "dodge")`

Question 3

```
ggplot(data=data, aes(x=smoking, y=percent))+  
  geom_bar(aes(fill = lung_cancer), stat = "identity", position = "dodge")
```

Percent of given
smoking status, if they
have lung cancer or not

